Assignment–3

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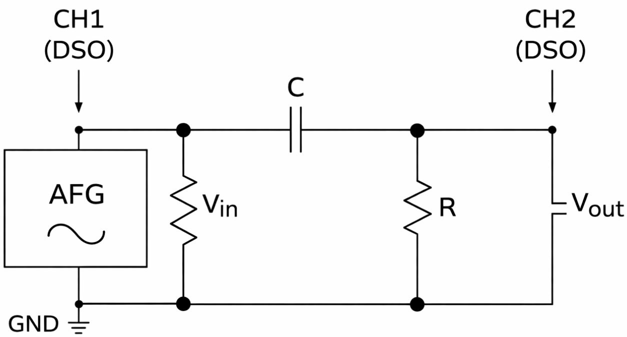
Part A: Step Input (Single Event)

Our aim is study the time domain response of an CR circuit for a step input , capture the charging and discharging waveforms determine the time constant experimentally and verify it using theoretical calculations and simulations

The following component values were chosen for the experiment:

- Resistance, $R = 1\,\text{k}\Omega = 1000\,\Omega$
- Capacitance, $C = 1\,\mu\text{F} = 1 \times 10^{-6}\,\text{F}$
- Input step voltage, $V = 5\,\text{V}$

CR Circuit diagram:



Time Constant Calculation

The time constant of an RC circuit is given by:

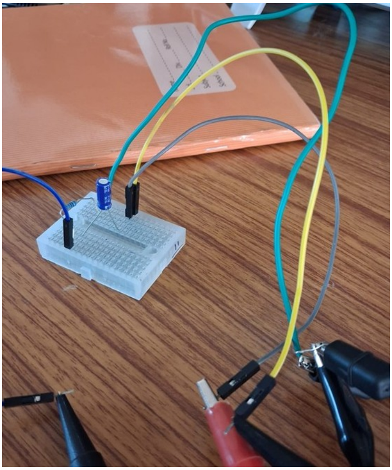
$$\tau = RC$$

$$\tau = 1000 \times 1 \times 10^{-6}$$

$$\tau = 1 \times 10^{-3}\text{ s} = 1\text{ ms}$$

In a CR circuit, the voltage is measured across the resistor. When a step input is applied, the capacitor initially acts as a short circuit. Hence, the current is maximum and the voltage across the resistor is also maximum.

CR Circuit Connections



Resistor Voltage During Capacitor Discharging

When the input changes, the capacitor starts discharging through the resistor. Since the capacitor voltage cannot change instantly, a large current flows initially, so the voltage across the resistor is maximum with opposite polarity.
As time increases, the current decreases and the resistor voltage decays exponentially:

$$V_R(t) = -Ve^{-t/RC}$$

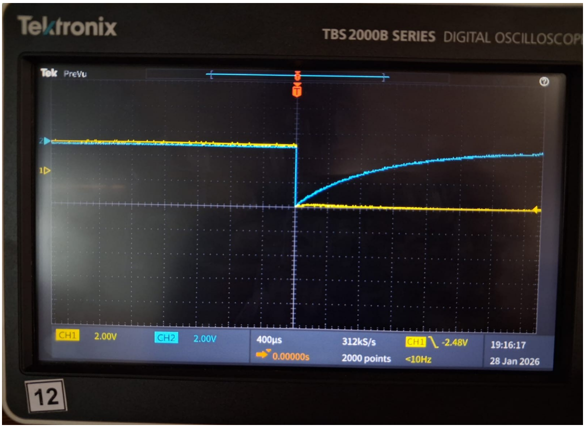
At $t = \tau = RC$,

$$V_R(\tau) = -Ve^{-1}$$

$$V_R(\tau) = -0.368V$$

For $V = 5\text{ V}$,

$$V_R(\tau) = -5 \times 0.368 = -1.84\text{ V}$$



Resistor Voltage During Capacitor Charging

As the capacitor charges, the current decreases exponentially. Therefore, the voltage across the resistor also decreases exponentially.
The voltage across the resistor is given by

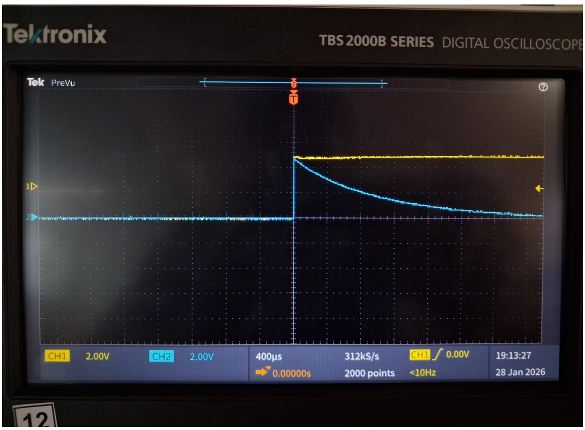
$$V_R(t) = Ve^{-t/RC}$$

At $t = \tau = RC$,

$$V_R(\tau) = Ve^{-1} = 0.368V$$

For $V = 5\,\text{V}$,

$$V_R(\tau) = 5 \times 0.368 = 1.84\,\text{V}$$



Percentage Error Calculation

From the above DSO waveforms, the experimental time constant is found to be

$$\tau_{\text{exp}} = 1.04\,\text{ms}$$

The percentage error is calculated as:

$$\text{Error} = \left(\frac{|\tau_{\text{theo}} - \tau_{\text{exp}}|}{\tau_{\text{theo}}}\right) \times 100$$

From the above calculations,

$$\tau_{\text{theo}} = 1\,\text{ms}$$

$$\text{Error} = \left(\frac{|1.00 - 1.04|}{1.00}\right) \times 100$$

$$\text{Error observed} = 4\%$$

The observed error of 4% is due to slight variations in component values and measurement errors in the oscilloscope. The function generator also introduces minor loading effects on the circuit.
Despite this small error, the experimental value closely matches the theoretical value. Hence, the behavior of the CR circuit is verified.

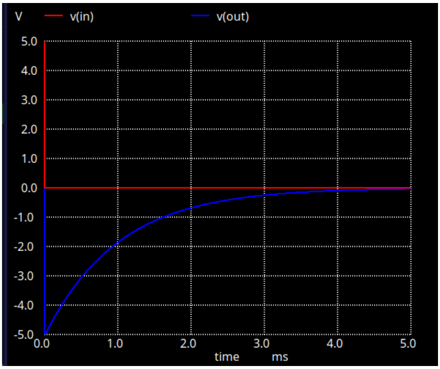
Simulation

To verify the theoretical calculations, the CR circuit was simulated using NGspice.

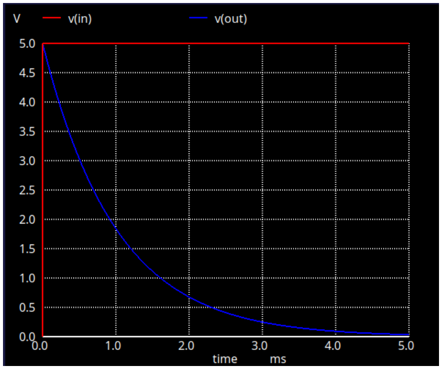
NGspice Netlist

```
* CR Discharging
V1 in 0 PULSE(5 0 0 1n 1n 10m 20m)
C1 in out 1uF
R1 out 0 1k

.tran 0.01ms 5ms
.control
run
plot v(in) v(out)
.endc
.end
```

```
* CR Charging
V1 in 0 PULSE(0 5 0 1n 1n 10m 20m)
C1 in out 1uF
R1 out 0 1k

.tran 1us 5ms
.control
run
plot v(in) v(out)
.endc
.end
```



DIscharging



charging

Comparison table:

Parameter	Theoretical	Experimental	Simulation
Time constant, τ (ms)	1.00	1.04	1.01
$V_R(\tau)$ during charging (V)	1.84	1.80	1.83
$V_R(\tau)$ during discharging (V)	-1.84	-1.78	-1.82

Conclusion

The transient response of the CR circuit was studied by measuring the voltage across the resistor. The resistor voltage decayed exponentially during both charging and discharging. At the time constant $\tau \approx 1$ ms, the voltage reached about 36.8% of its initial value. The experimental results closely matched the theoretical values.

Part B: Pulse Input (Single Event)

A single pulse of 5 V amplitude was applied to the CR circuit. Pulse widths of $200\,\mu s$, $500\,\mu s$, and $800\,\mu s$ were tested. The repetition period was kept large enough to allow complete discharge of the capacitor.

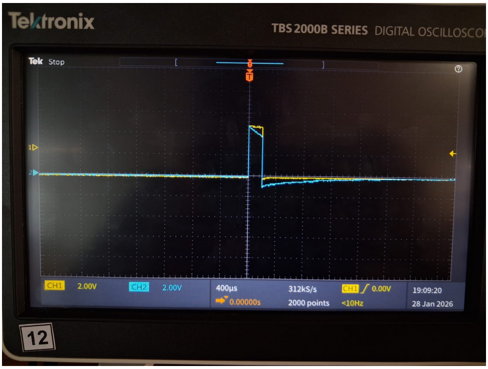
Theoretical Analysis:
The voltage across the resistor in a CR circuit is given by:

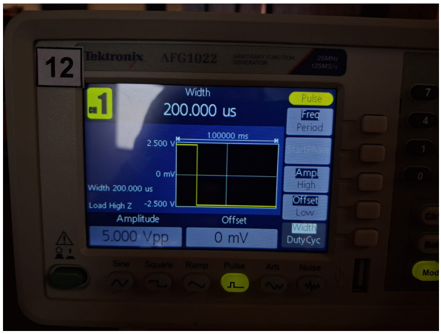
$$V_R(t) = \pm V e^{-t/RC}$$

- **Pulse width =** $200\,\mu s$

$$V_{\text{fall}} = -5e^{-0.2} = -5(0.819) = -4.09\ \text{V}$$

A sharp positive spike of +5 V is observed at the rising edge followed by an exponential decay. A large negative spike of −4.09 V appears at the falling edge since the capacitor does not fully discharge during the pulse.

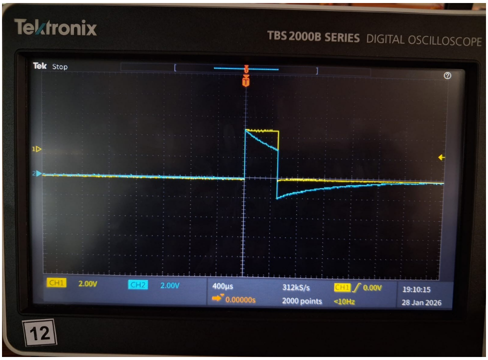


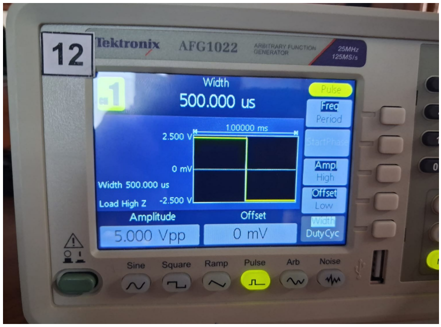


- **Pulse width =** $500\,\mu s$

$$V_{\text{fall}} = -5e^{-0.5} = -5(0.607) = -3.03\ \text{V}$$

A positive spike is observed at the rising edge followed by exponential decay. The negative spike of −3.03 V is smaller than that of $200\,\mu s$, indicating partial discharge of the capacitor during the pulse.

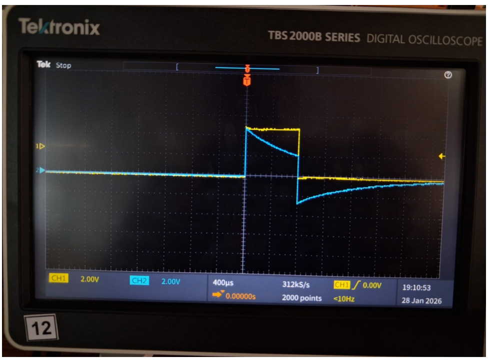


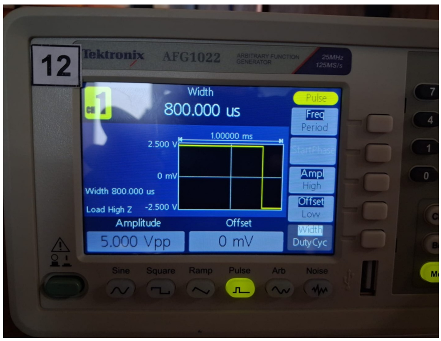


- **Pulse width =** $800\,\mu s$

$$V_{\text{fall}} = -5e^{-0.8} = -5(0.449) = -2.24\,\text{V}$$

The rising edge produces a positive spike, and the exponential decay is almost complete before the falling edge. The negative spike of -2.24 V is much smaller due to greater discharge of the capacitor during the longer pulse.





Simulation

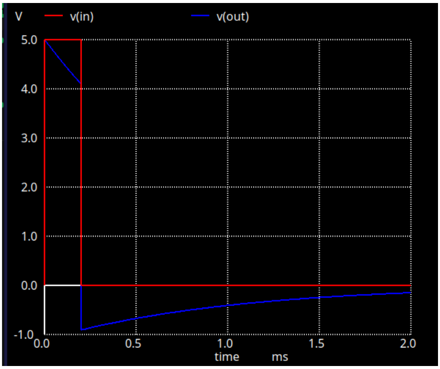
To verify the theoretical calculations, the CR circuit was simulated using NGspice.

NGspice Netlist

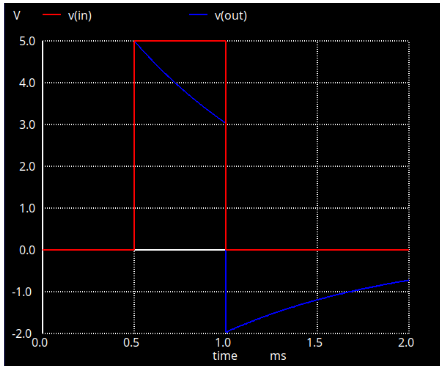
```
* CR Circuit - Single Pulse 200us
V1 in 0 PWL(0 0 1u 0 1.001u 5 201u 5 201.001u 0)
C1 in out 1uF
R1 out 0 1k

.tran 1u 2m
.control
run
plot v(in) v(out)
.endc
.end
```

```
* CR Circuit - Single Pulse 500us
V1 in 0 PWL(0 0 500u 0 500.001u 5 1000u 5 1000.001u 0)
C1 in out 1uF
R1 out 0 1k

.tran 1u 2m
.control
run
plot v(in) v(out)
.endc
.end
```

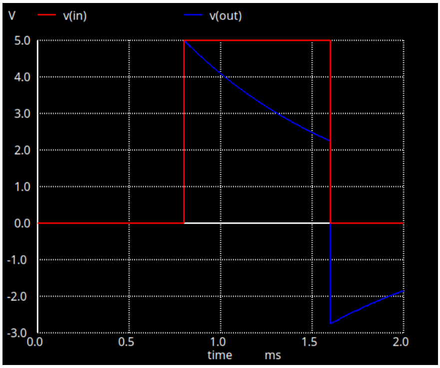
```
* CR Circuit - Single Pulse 800us
V1 in 0 PWL(0 0 800u 0 800.001u 5 1600u 5 1600.001u 0)
C1 in out 1uF
R1 out 0 1k

.tran 1u 2m
.control
run
plot v(in) v(out)
.endc
.end
```



$200\,\mu s$



$500\,\mu s$



$800\,\mu s$

Comparison Table

Pulse Width	Theoretical V_{fall} (V)	Experimental V_{fall} (V)	Simulation V_{fall} (V)
$200\,\mu s$	-4.09	≈ -4.0	-4.1
$500\,\mu s$	-3.03	≈ -3.0	-3.0
$800\,\mu s$	-2.24	≈ -2.2	-2.3

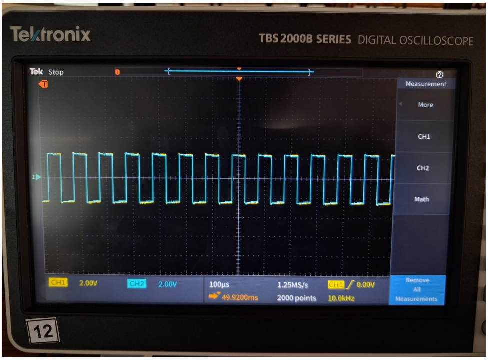
Conclusion:
The response of the CR circuit depends on the pulse width. For shorter pulse widths, the capacitor does not discharge fully, resulting in a larger negative spike at the falling edge. As the pulse width increases, the capacitor discharges more during the pulse, and the magnitude of the negative spike decreases. The experimental results agree well with the theoretical and simulation values.

Part C: Square-Wave Input (Transient and Steady-State Response)

This experiment analyzes the response of a **CR circuit** to a square-wave input for different relationships between the time constant (RC) and the input time period (T). The observed output waveforms are compared with theoretical predictions and NGSPICE simulation results.

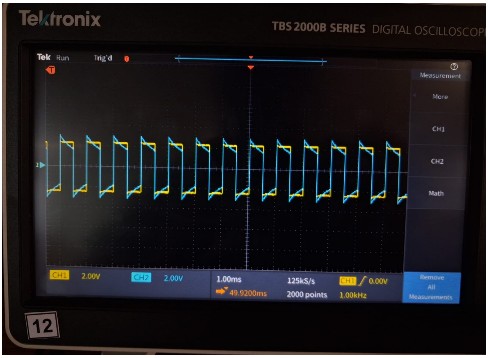
Case 1: $RC \gg T$

Input condition: $RC = 1$ ms, $f = 10$ kHz, $T = 0.1$ ms.
The output waveform consists of narrow spikes at the edges of the input signal. Since $T \ll RC$, the capacitor does not get enough time to discharge fully, causing reduced amplitude and a waveform that appears nearly continuous.



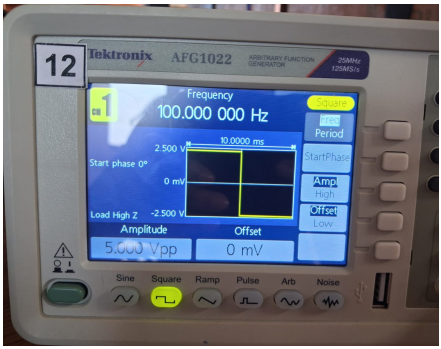
Case 2: $RC = T$

Input condition: $RC = 1$ ms, $f = 1$ kHz, $T = 1$ ms.
When the time period is approximately equal to the time constant, the output waveform shows distinct positive and negative pulses with exponential decay. In this case, both transient and steady-state responses are clearly observed.



Case 3: $RC \ll T$

Input condition: $RC = 1$ ms, $f = 100$ Hz, $T = 10$ ms.
When the time period is much larger than the time constant, sharp and narrow spikes appear at the rising and falling edges of the input signal. The capacitor fully charges and discharges, and the circuit behaves like a differentiator.



Simulation

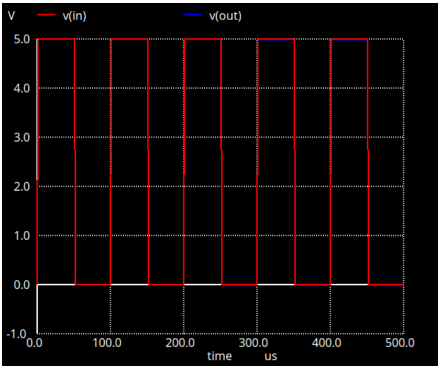
NGSPICE simulations were performed using a square-wave voltage source with different frequencies corresponding to each case. The output voltage across the capacitor was observed and compared with theoretical and experimental results.

NGspice Netlist

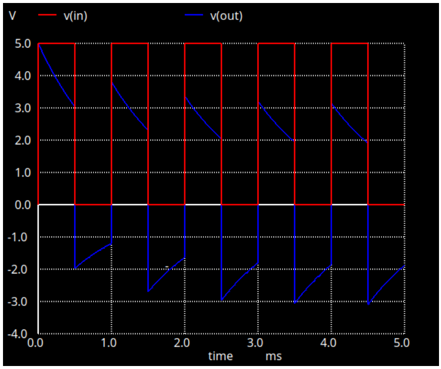
```
* CR response for RC >> T (10 kHz square-wave input)

V1 in 0 PULSE(0 5 0 1u 1u 50u 100u)
C1 in out 1u
R1 out 0 1k

.tran 0.1u 500u
.control
run
plot v(in) v(out)
.endc
.end
```

```
* CR response for RC = T (1 kHz square-wave input)
V1 in 0 PULSE(0 5 0 1u 1u 0.5m 1m)
C1 in out 1u
R1 out 0 1k


.tran 1u 5m
.control
run
plot v(in) v(out)
.endc
.end
```

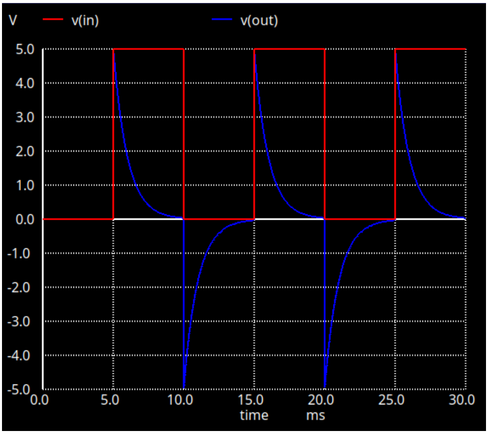
```
 * CR response for RC << T (100 Hz square wave)

V1 in 0 PULSE(0 5 5m 1n 1n 5m 10m)
C1 in out 1uF
R1 out 0 1k

.tran 0.01m 30m
.control
run
plot v(in) v(out)
.endc
.end
```



$RC \gg T$



$RC \approx T$



$RC \ll T$

Frequency	**Time Period (ms)**	**RC vs T**	**Output Nature**
10 kHz	0.10	$RC \gg T$	Very small, closely spaced spikes
1 kHz	1.00	$RC \approx T$	Distinct positive and negative pulses with exponential decay
100 Hz	10.0	$RC \ll T$	Sharp spikes at transitions

Part D: Frequency Response of RC Circuit

Theory

For an RC circuit with the output taken across the capacitor, the circuit behaves as a low-pass filter. The magnitude of the transfer function is given by:

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

The cut-off frequency is defined as:

$$f_c = \frac{1}{2\pi RC}$$

At cut-off frequency, the gain is:

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{2}} \approx 0.707$$

Calculations

Gain is calculated using:

$$\text{Gain} = \frac{V_{out}}{V_{in}}$$

Gain in decibels is calculated using:

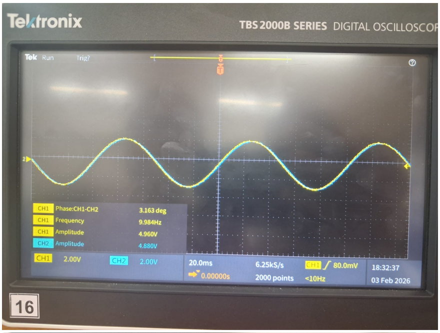
$$\text{Gain (dB)} = 20 \log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

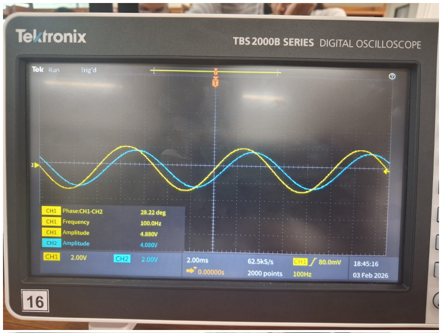
Theoretical cut-off frequency:

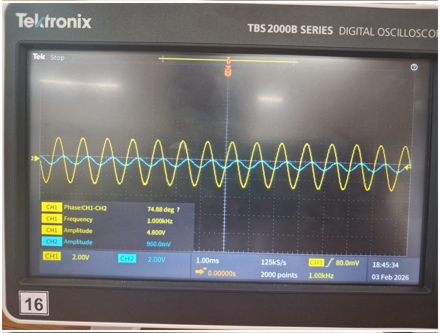
$$f_c = \frac{1}{2\pi RC}$$

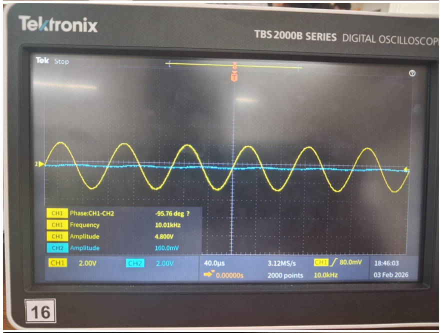
Observations

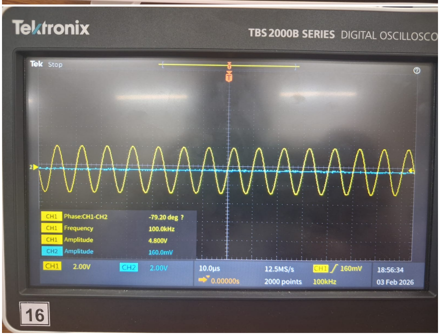
Frequency (Hz)	V_{in} (V)	V_{out} (V)	Gain $\lvert V_{out}/V_{in}\rvert$	Gain (dB)	Phase (deg)
10	4.96	4.88	0.98	-0.17	3.16
100	4.80	4.24	0.88	-1.11	28.22
1 k	4.72	0.96	0.20	-13.98	74.88
10 k	4.88	0.16	0.03	-29.7	95.76
100 k	4.96	0.16	0.03	-29.7	79.20
1 M	5.04	0.16	0.03	-29.7	–

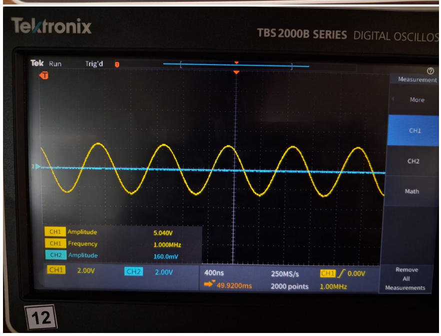






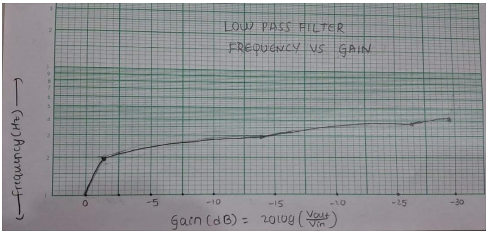


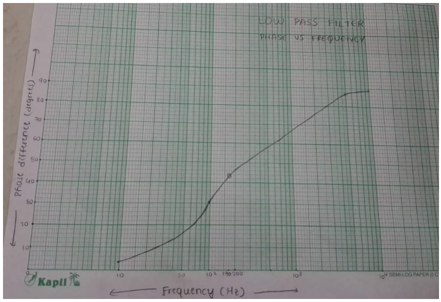




Frequency Response Plot

A semi-logarithmic plot of Gain (dB) versus Frequency was plotted. The gain is maximum at low frequencies and decreases as the frequency increases, showing attenuation at higher frequencies.





Cut-off Frequency

The cut-off frequency is identified as the frequency at which the gain is -3 dB. From the magnitude response plot, the cut-off frequency is observed to be approximately:

$$f_c \approx 100\,\text{Hz}$$

Slope of the Magnitude Response

Above the cut-off frequency, the slope of the magnitude response is approximately:

$$-20\,\text{dB/decade}$$

Below the cut-off frequency, the gain remains nearly constant.

Result

The frequency response of the given RC circuit was obtained experimentally. The circuit allows low-frequency signals to pass and attenuates high-frequency signals.

Hence, the RC circuit behaves as a Low-Pass Filter.

Part E: Frequency Response of CR Circuit (High-Pass)

Theory

For a CR circuit with the output taken across the resistor, the circuit behaves as a high-pass filter. At low frequencies, the capacitive reactance is high and the output voltage is small. At high frequencies, the capacitive reactance decreases and the output voltage approaches the input voltage.

The magnitude of the transfer function of a high-pass RC circuit is given by:

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

The cut-off frequency is defined as:

$$f_c = \frac{1}{2\pi RC}$$

At the cut-off frequency, the gain is:

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{2}} \approx 0.707$$

Calculations

Gain is calculated using:

$$\text{Gain} = \frac{V_{out}}{V_{in}}$$

Gain in decibels is calculated using:

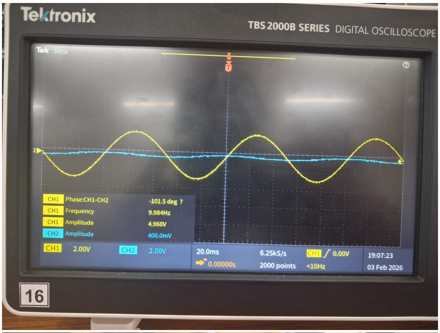
$$\text{Gain (dB)} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

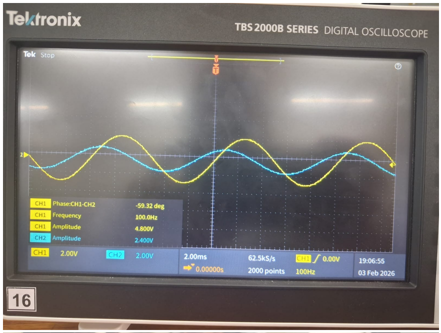
Theoretical cut-off frequency:

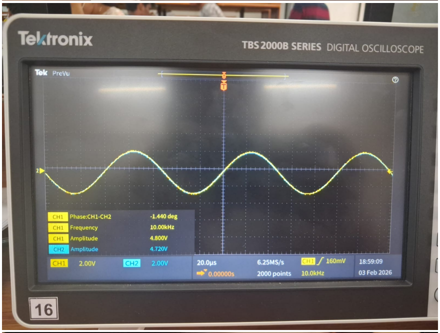
$$f_c = \frac{1}{2\pi RC}$$

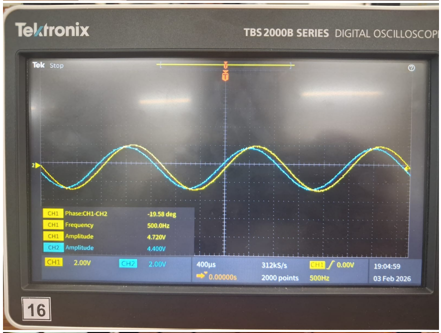
Observations

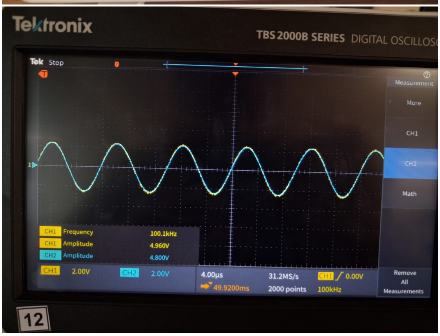
Frequency	V_{in} (V)	V_{out} (V)	Gain	Gain (dB)	Phase Angle (°)
10 Hz	4.96	0.48	0.097	-20.3	-101.5
100 Hz	4.88	2.48	0.508	-5.88	-59.3
1 kHz	4.80	4.56	0.95	-0.45	≈ -15
10 kHz	4.88	4.80	0.98	-0.17	-1.44
100 kHz	4.96	4.80	0.97	-0.26	≈ -0.5
1 MHz	5.12	4.96	0.97	-0.26	≈ 0

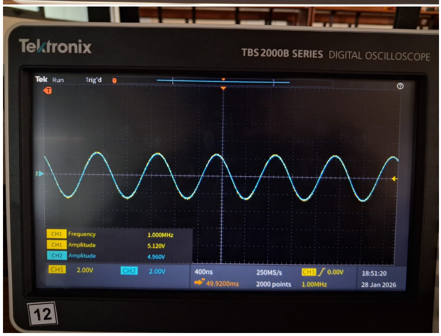






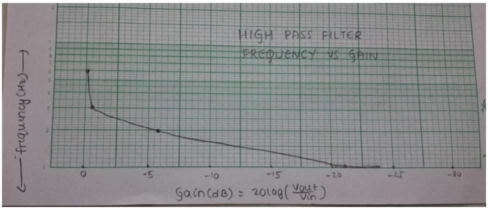


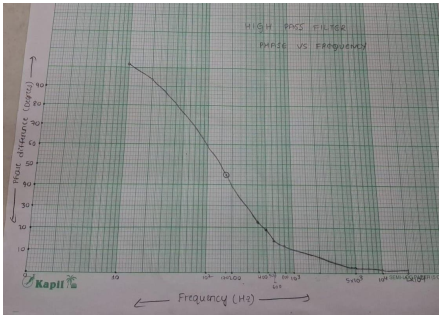




Frequency Response Plot

A semi-logarithmic plot of Gain (dB) versus Frequency was plotted. The gain is low at low frequencies and increases with frequency, approaching 0 dB at higher frequencies.





Cut-off Frequency

The cut-off frequency is identified as the frequency at which the gain is -3 dB. From the magnitude response plot, the cut-off frequency is observed to be approximately:

$$f_c \approx 100\,\text{Hz}$$

Slope of the Magnitude Response

Below the cut-off frequency, the slope of the magnitude response is approximately:

$$+20\,\text{dB/decade}$$

Above the cut-off frequency, the gain remains nearly constant.

Result

The frequency response of the given CR circuit was obtained experimentally. The circuit attenuates low-frequency signals and allows high-frequency signals to pass.

$\boxed{\text{Hence, the CR circuit behaves as a High-Pass Filter.}}$